Bid Optimization for Search Advertising in Non-stationary Environment

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Abstract. To achieve high performance in search advertising campaigns, it is crucial to set the bid value appropriately. However, optimal bidding is challenging due to the high uncertainty in ad performance, which is influenced by various factors besides the bid value. In this paper, we classify the influence on the performance into three types: stationary factors, non-stationary deterministic factors, and non-stationary non-deterministic factors. Considering that these factors change the campaign environment, we propose a heuristic optimization algorithm that constructs a prediction model for each known campaign environment and utilizes past results directly to determine the optimal bid value. The proposed method allows for applying different Cost-Per-Click (CPC) constraints specific to each campaign environment, in addition to the overall campaign CPC constraint. We validated the effectiveness of the proposed method using a simulator and demonstrated that it achieves a large number of ad clicks within the constraints in the non-stationary environment.

Keywords: Bid optimization · Search advertising · Non-stationary environment.

1 Introduction

Search advertising, also known as keyword advertising or listing advertising, is an online advertising displayed on a search results page based on the keywords a user enters into a search engine. It is one of the most widely used online advertising forms, and its market has continued to grow in recent years [5].

Search advertising allows for flexible adjustment of settings, including ad content, target users, target keywords, campaign duration, budget, and bid value. By analyzing past results and adjusting these settings, the advertiser can achieve high performance. In this paper, high performance is defined as maximizing the total number of clicks throughout the campaign, subject to two constraints: total costs being below the budget and Cost-Per-Click (CPC) being below a predetermined target value.

However, it is challenging to adjust settings appropriately. The main reasons for this difficulty are the limited information regarding past results available to advertisers and the high uncertainty in ad performance. The information available to the advertiser is limited to the historical settings and the resulting performance. The scarcity of data hinders the accurate prediction of performance changes under modified settings. Furthermore, the performance is influenced by various factors outside of the parameters set by the advertiser. Consequently, results can fluctuate significantly even with the same settings, and the best setting is constantly changing.

In this paper, we propose a method for automatically determining the bid value for search advertising, where bids can be set at the keyword level, to achieve high performance. The main contributions of this paper are threefold:

- We classify the influences on ad performance into three types: stationary factors, non-stationary deterministic factors, and non-stationary non-deterministic factors, and we determine the optimal bid value by considering the effect of each factor. No previous research has been found that categorizes the uncertainty factor in performance and optimizes bid value.
- We determine the bid value such that ad performance satisfies multiple CPC targets set within the campaign period in accordance with the non-stationary deterministic factors. Although major advertising platforms in recent years come standard with automated bidding features, they cannot handle such complex settings.
- We validated the effectiveness of the proposed method through ad operation simulations. The results show that high performance can be achieved under the various factors considered in this study.

2 Problem Setting

This section explains the search advertising system, which is the focus of this paper, and the three types of factors that influence ad performance, and then formulates the bid optimization problem for search advertising.

2.1 Search Advertising

Search advertising is facilitated by a connection between multiple advertisers and users through an ad delivery platform. When a user searches for a keyword on a search engine, the platform selects an ad via its auction system from those associated with that keyword. Although the auction system is private and varies by platform, the evaluation considers a quality score, which is assessed based on elements such as relevance of the ad and the keyword, and estimated Click-Through Rate (CTR), in addition to the bid value set by the advertiser for each keyword. Displaying an ad to a user is called an impression. Subsequently, the user may click on the displayed ad. Furthermore, after clicking the ad, the user may take a final desired action for the advertiser, such as a product purchase, membership registration, or document request. This action is called a conversion. In this paper, we assume the Pay-Per-Click (PPC) model, which is adopted by major platforms. Under PPC, the advertiser is charged the ad cost only when a user clicks the ad.

The adjustable parameters for an advertiser on the search advertising platform considered in this paper are the target keywords, campaign duration, total budget, and the bid value for each keyword. The keywords, duration, and budget are set before the campaign starts and cannot be modified during the campaign. On the other hand, the bid can be freely changed during the campaign. We assume the duration is set at the monthly level and the bid is adjusted daily based on practical ad operation procedures. If the total budget is exhausted during the campaign period, ad delivery will be suspended thereafter.

The performance results available for an advertiser on the platform considered in this study are the number of impressions, clicks, conversions, and costs for each keyword. Note that information such as keyword search volume, ad quality score, or competitor information is not included. Since the results are available daily, advertisers can analyze them up to the previous day and adjust the bid value for the current day.

2.2 Factors Affecting Advertising Operations

As mentioned in Section 1, ad performance constantly fluctuates, even under the same settings, due to various influences beyond the parameters set by the advertiser. We classify and consider these influences as three types of factors: stationary, non-stationary deterministic, and non-stationary non-deterministic.

Stationary factors These factors continuously affect the campaign throughout the entire period. Examples include the uncertainty in user and competitor behavior. The low CTR of search advertising is a major cause of instability, particularly in the resulting number of clicks and costs.

Non-stationary deterministic factors These factors change during the ad campaign, but their changes can be anticipated before the campaign starts. Examples include holidays, sales, and related ad delivery on other media. These factors often affect user behavior, such as keyword search volume and the number of ad clicks.

Non-stationary non-deterministic factors These factors change during the campaign, and their changes cannot be captured until it is observed. Examples include information dissemination through third-party media like SNS or TV, and the entry of new competitor advertisers. Media coverage affects user search volume and clicks, while new entrants influence competitive bid value.

In this paper, we refer to the totality of influences on campaign results originating from factors outside those set by the advertiser as the operational environment. Furthermore, we define a non-stationary environment as one where the operational environment changes during the campaign period. In this context, stationary factors are independent of changes in the operational environment, while non-stationary deterministic factors and non-stationary non-deterministic factors are factors that cause changes in the operational environment. Notably, changes in the operational environment due to non-stationary deterministic factors can be identified before the start of an advertising campaign.

2.3 Formulation

The notations are summarized in Table 1. Note that the true functions are unknown in practice; the advertiser needs to predict those functions based only on the results of clicks and costs obtained from the bid value set by themselves.

Table 1: Notations

Symbol	Type	Description
$\mathcal{A}$	set	non-stationary deterministic factors
$\mathcal{L}_a$	set	labels that factor $a \in \mathcal{A}$ can take
$\mathcal{L}_\mathcal{S}$	set	all possible combinations of labels that each factor in $\mathcal{S} \subseteq \mathcal{A}$ can take ($\prod_{s \in \mathcal{S}} \mathcal{L}_s$)
$\mathcal{T}$	set	days included in the campaign period
$\mathcal{T}^{(\mathcal{S},\ell_\mathcal{S})}$	set	days included in the environment that $\mathcal{S}$ takes $\ell_\mathcal{S} \in \mathcal{L}_\mathcal{S}$
$\mathcal{W}$	set	target keywords
$\mathcal{X}$	set	bid values
B	constant	budget for the entire period
K	constant	CPC target for the entire period
$K^{(\mathcal{S},\ell_\mathcal{S})}$	constant	CPC target for the environment that $\mathcal{S}$ takes $\ell_\mathcal{S}$
$x_{t,i}$	variable	bid value for keyword $i \in \mathcal{W}$ on day $t \in \mathcal{T}$
$v_{t,i}(\cdot)$	function	number of clicks for keyword i on day t
$c_{t,i}(\cdot)$	function	ad costs for keyword i on day t

The problem is to determine the bid value that maximizes total clicks while satisfying the constraints that total costs over the campaign duration remain below the budget and that CPC remains below the target value. We address the problem of setting individual CPC targets for each known change of operational environment besides the overall CPC target for the entire period, to avoid CPC deterioration in specific operational environments. Therefore, the optimization problem is formulated as follows:

$$
\begin{aligned}
\max_{x_{t,i} \in \mathcal{X}} \quad & \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{W}} v_{t,i}(x_{t,i}) \\
\text{s.t.} \quad & \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{W}} c_{t,i}(x_{t,i}) \leq B, \\
& \frac{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{W}} c_{t,i}(x_{t,i})}{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{W}} v_{t,i}(x_{t,i})} \leq K, \\
& \frac{\sum_{t \in \mathcal{T}^{(\mathcal{S},\ell_\mathcal{S})}} \sum_{i \in \mathcal{W}} c_{t,i}(x_{t,i})}{\sum_{t \in \mathcal{T}^{(\mathcal{S},\ell_\mathcal{S})}} \sum_{i \in \mathcal{W}} v_{t,i}(x_{t,i})} \leq K^{(\mathcal{S},\ell_\mathcal{S})}, \ \forall \mathcal{S} \subseteq \mathcal{A}, \ \forall \ell_\mathcal{S} \in \mathcal{L}_\mathcal{S}.
\end{aligned}
\tag{1}
$$

When it is unnecessary to set a CPC target $K^{(\mathcal{S},\ell_{\mathcal{S}})}$, simply set it to a sufficiently large value.

3 Related Work

This section presents previous research on bid optimization for online advertising that addresses problem settings similar to our study, not limited to search advertising, while explaining the challenges in applying it to this problem setting. Each research differs in its problem setting regarding the objective, the number of target keywords or ad slots, and the system related to the advertising platform. However, only Majima et al. [9] target the same search advertising system as our research.

First, many studies target platforms where performance results data can be obtained and bid value updated per search, i.e., per auction. The bandit problem framework and deep learning methods used in these studies are unsuitable for the problem setting in this paper, where bid value is updated daily, resulting in fewer rounds. Abhishek & Hosanagar [1], which deal with daily updates, treats the bid optimization as a stochastic optimization problem, while Majima et al. [9] treat it as a hierarchical Bayesian model and heuristic bandit.

Second, no study has addressed CPC constraints specific to operational environments, and not all studies address CPC constraints throughout the entire period. In Yang et al. [13], which addresses CPC constraint over the entire period, the parameters included in the formula representing the optimal bid value are estimated using model predictive control. Liang et al. [7] treat the bid optimization with Return-On-Investment constraint as an online optimization problem, while Vijayan et al. [12] treat it with Return-On-Spend constraint as a bandit problem.

Third, most studies do not address a non-stationary environment, and even those that do address it do not explicitly consider the factors involved. Zhao et al. [14], which addresses changes in operational environments, deal with an operational environment corresponding to the stationary environment discussed in Section 2.2 of this paper, where performance results stabilize when time is aggregated. In Liang et al. [7], multiple patterns of operational environment changes are assumed, and theoretical analyses are conducted for each.

4 Proposed Method

This section describes the proposed method of this paper. The proposed method adopts a framework that sequentially repeats prediction and optimization [10,9]: it predicts functions of clicks and costs based on campaign results up to the previous day, and then solves an optimization problem to determine the bid value for the current day using these functions. In this paper, we define the roles corresponding to the update subroutine and exploration subroutine in [10,9] as the prediction phase and the optimize subroutine as the optimization phase, constituting a two-stage process.

Here, we define the term "unit environment" for the purpose of explaining the proposed method. A unit environment refers to an operational environment that cannot be further divided into multiple operational environments. It is represented by assigning labels to all non-stationary deterministic factors. Each day during the campaign period is assigned to exactly one unit environment. That is, the label for day t corresponds to one of the unit environments.

4.1 Prediction Phase

In the prediction phase, based on campaign results up to the previous day, we predict the functions for the number of clicks and costs used for optimization. The two challenges, the limited information available to advertisers and the high uncertainty in ad performance, can lead to reduced forecast accuracy. In particular, since daily performance results fluctuate due to the influence of stationary factors, predicting the function for each day is difficult. Therefore, we predict the function for each unit environment, categorized by non-stationary deterministic factors, and assign it to the corresponding day.

To address the challenge of limited available information, we train a common prediction model for clicks or costs across all unit environments and all keywords. The prediction model uses XGBoost [3] with bid value, keyword ID, and operation environment label as features. Furthermore, we introduce privileged information learning [8] as a framework to incorporate features that are available during model training but not during prediction. We add impressions and costs as privileged information for the click prediction model, while we add impressions and clicks as privileged information for the cost prediction model. For the teacher model, a simple model that avoids overfitting is employed. Considering multicollinearity among impressions, clicks, and costs, we adopted ridge regression for the teacher model.

To address the high uncertainty in performance, we introduce quantile regression [11] and monotonic regression [6]. By predicting quantiles below 0.5 for clicks and quantiles above 0.5 for costs, we achieve conservative ad operations where even if predictions are inaccurate, the possibility of constraint violations is low. Furthermore, as bid value increases, the auction win rate rises, leading to higher clicks and costs. Therefore, assuming a monotonically increasing function is expected to improve forecast accuracy.

4.2 Optimization Phase

During the optimization phase, the optimal bid value to set is determined using predicted functions of the clicks and costs.

Consider Problem (1) for day t. At this point, the values of clicks and costs up to day $t-1$ are fixed because the ad campaign has concluded. Therefore, the clicks and costs obtained up to day $t-1$ are treated as constants. This approach addresses the uncertainty in advertising caused by stationary factors and non-stationary non-deterministic factors. Furthermore, the functions for the number of clicks and costs from day t onwards are predicted separately for each unit environment and keyword. Corresponding to these predictions, the bid values from day t onwards are also treated as variables specific to each unit environment and keyword. This allows explicit consideration of non-stationary deterministic factors. This formulation in the optimization phase is a natural extension of the formulation in Majima et al. [9] to accommodate non-stationary environments and CPC constraints.

Let $\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}$ be the set of days contained in $\mathcal{T}^{(\mathcal{A},\ell_\mathcal{A})}$ and starting from day t, and $V_{t-1}^{(\mathcal{S},\ell_\mathcal{S})}$, $C_{t-1}^{(\mathcal{S},\ell_\mathcal{S})}$ be the constants representing the total number of clicks or costs on days up to day $t-1$ that belong to the unit environment labeled $\ell_\mathcal{A}$. Also, let $x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}$ denote the variable representing the bid value on keyword i in unit environment $\ell_\mathcal{A}$, and $\hat{v}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(\cdot)$, $\hat{c}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(\cdot)$ denote the functions of the number of clicks or costs on keyword i in unit environment $\ell_\mathcal{A}$, predicted during the prediction phase. The optimization problem to solve in day t is formulated as follows:

$$\begin{aligned}
\max_{x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}\in\mathcal{X}} \quad & \sum_{\ell_\mathcal{A}\in\mathcal{L}_\mathcal{A}} \left(V_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{v}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right) \\
\text{s.t.} \quad & \sum_{\ell_\mathcal{A}\in\mathcal{L}_\mathcal{A}} \left(C_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{c}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right) \le B, \\
& \frac{\sum_{\ell_\mathcal{A}\in\mathcal{L}_\mathcal{A}} \left(C_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{c}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right)}{\sum_{\ell_\mathcal{A}\in\mathcal{L}_\mathcal{A}} \left(V_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{v}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right)} \le K, \\
& \frac{\sum_{\ell_\mathcal{A}\supseteq\ell_\mathcal{S}} \left(C_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{c}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right)}{\sum_{\ell_\mathcal{A}\supseteq\ell_\mathcal{S}} \left(V_{t-1}^{(\mathcal{A},\ell_\mathcal{A})} + |\mathcal{T}_t^{(\mathcal{A},\ell_\mathcal{A})}| \sum_{i\in\mathcal{W}} \hat{v}_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}(x_{t,i}^{(\mathcal{A},\ell_\mathcal{A})}) \right)} \le K^{(\mathcal{S},\ell_\mathcal{S})}.
\end{aligned} \tag{2}$$

Problem (2) is a nonlinear and nonconvex optimization problem, making it difficult to solve. Therefore, similar to Majima et al. [9], we convert it to 0–1 integer linear programming by utilizing the fact that the bid value is a discrete variable, and solve it.

5 Numerical Experiments

This section describes the ad operation simulations conducted to verify the effectiveness of the proposed method. The simulations verified whether high click counts could be achieved while satisfying budget constraints and CPC constraints when various factors affecting ad performance, as described in Section 2.2, were present during the operation.

5.1 Setup

We simulated ad operations using the sponsored search simulator employed in the numerical experiments of Majima et al. [9]. For the parameters set in advance by the advertisers, the target keywords consisted of five high-search-volume keywords, five low-search-volume, high-CTR keywords, and five low-search-volume, low-CTR keywords, for a total of 15 keywords. The operation period was set to 30 days ($t = 1, \dots, 30$), corresponding to the minimum configurable unit of one month. It was assumed that 30 days of preliminary test data using random bid values exist. The budget was set to 50% of the value that maximizes total cost. Target CPC values were set to 0.6 and 0.8 times the value that maximizes CPC, and we compared performance under each target. Furthermore, when the CPC constraint was 0.6, we also compared it to a case without CPC constraints for each unit environment.

This numerical experiment simulated non-stationary environments by varying the parameters for each keyword's average search volume, CTR, and average competitor bid daily. The settings of these non-stationary factors and their effects are summarized in Table 2, where each factor is expressed as a scale relative to the simulator's normal parameters. For the normal parameters, we used values similar to those in [9], based on actual ad campaign log data. Combining these factors allows us to represent realistic non-stationary conditions.

Table 2: Parameters for non-stationary factors

Factor	Deterministic	Volume	CTR	Competitor's bid
Weekday	YES	$\times 1.00$	$\times 1.00$	$\times 1.00$
Weekday & Sale	YES	$\times 1.50$	$\times 1.50$	$\times 1.20$
Holiday	YES	$\times 1.20$	$\times 1.00$	$\times 1.10$
Holiday & Sale	YES	$\times 1.80$	$\times 1.50$	$\times 1.30$
Media	NO	$\times 2.00$	$\times 1.50$	$\times 1.00$
New-entry	NO	$\times 1.00$	$\times 1.00$	$\times 1.05$

We assumed a scenario in which a pre-scheduled sale period overlapped with operations that account for weekdays and holidays, user activity increased due to media exposure after the sale, and this change triggered new competitor entries. In the one-month ad campaign starting on Monday, a sale occurred during the second week ($t = 7, \dots, 13$). The ad performance was influenced by media exposure during the two days immediately following the sale ($t = 14, 15$), and by new-entry over the subsequent 14 days ($t = 16, \dots, 30$). Additionally, preliminary experiments were conducted for environments where only one non-stationary factor varied at a time.

Since this paper introduces a new problem formulation, existing methods cannot be directly applied for comparison. Therefore, we compared performance between setting the true optimal solution as the bid value (**True**) and setting the optimal solution obtained by dividing the day into segments without considering past results as the bid (**Naive**). **True** uses the offline optimal solution to (1), assuming full knowledge of the true click and cost functions. This benchmark is not feasible in real operations because those functions are unknown. **Naive** maximizes clicks on day t under two constraints: a daily budget obtained by evenly dividing the total budget across the campaign period, and a CPC constraint defined as the stricter of the overall CPC target or the CPC target for the unit environment labeled $\ell_{\mathcal{A}}$ on day t. Regarding the proposed method (**Proposal**), we also verified it when the functions in each unit environment were known, in order to confirm its predictive performance. Additionally, we compared results when introducing quantile regression to predict the 0.4 quantile of clicks and the 0.6 quantile of costs.

Finally, we describe the implementation of the methods. When predicting the functions of clicks and costs in the prediction phase, we employed Optuna [2] to adjust hyperparameters of the model. The tuning was performed once every 10 days by randomly splitting the observed data into training and validation sets in a $8:2$ ratio. When solving (2) to determine bids during the optimization phase, the mathematical optimization solver Gurobi Optimizer [4] is employed. During the optimization phase, the optimization problem sometimes became infeasible. In such cases, we solved the budget-constrained CPC minimization problem to set the bid value.

5.2 Results

We report and discuss the results for the experimental setting in Section 5.1, omitting the preliminary experiments. The evaluation metrics are total clicks, total costs, overall CPC, and CPC for each operational environment. Because ad performance is highly uncertain and outcomes vary with the simulator's seed, each setting is evaluated over five simulation runs.

The simulation results for the assumed scenarios are shown in Table 3; results of CPC for specific unit environments are omitted for brevity. Since the maximum clicks and constraint values vary across simulations, clicks are normalized to **True** (offline optimal) results, while costs and CPC are presented as ratios relative to their respective constraints, averaged over five runs. Table 4 reports the budget exhaustion rates and the frequency of infeasible optimization problems.

First, we compare the results under the 0.6 times CPC constraint, where the constraints were active. Except for **Proposal** that introduced quantile regression with known true functions, no operational results satisfying the CPC constraint were obtained. Also, even after introducing CPC constraints specific to each operational environment, the CPC for each environment did not improve. This is likely because it is difficult to accurately capture the characteristics of CPC, which is influenced by both clicks and costs. When violating a budget constraint, selecting a smaller bid value can resolve the violation. However, when violating a CPC constraint, it is unclear which bid value to select to resolve the violation. In fact, the trends and values of the predicted CPC function itself deviated significantly from the true CPC. Furthermore, because of the high uncertainty of clicks and cost for each day, even in methods where the true function for each unit environment was known, it was not guaranteed that operations could be conducted to satisfy the constraints. Due to the inaccuracy of predictions,

Table 3: Summary of campaign results 1

(a) Click

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	1.0000	1.0000	1.0000
Naive	-	NO	0.8749	0.8800	0.8010
Naive	-	YES	0.6746	0.6643	0.8458
Proposal	known	NO	0.7486	0.7550	0.9028
Proposal	known	YES	0.7200	0.7174	0.9029
Proposal	unknown	NO	0.6338	0.6059	0.8664
Proposal	unknown	YES	0.5994	0.5937	0.8704

(b) Cost

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	0.7799	0.7676	0.9999
Naive	-	NO	0.7864	0.7868	1.0000
Naive	-	YES	0.5391	0.5198	0.9787
Proposal	known	NO	0.5847	0.5820	0.9976
Proposal	known	YES	0.5585	0.5473	0.9959
Proposal	unknown	NO	0.5426	0.4882	0.9983
Proposal	unknown	YES	0.4724	0.4574	0.9956

(c) CPC

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	1.0000	0.9968	0.8156
Naive	-	NO	1.1506	1.1602	1.0118
Naive	-	YES	1.0240	1.0159	0.9443
Proposal	known	NO	1.0002	1.0011	0.9014
Proposal	known	YES	0.9948	0.9909	0.8997
Proposal	unknown	NO	1.0988	1.0468	0.9401
Proposal	unknown	YES	1.0107	1.0007	0.9331

(d) CPC (Holiday)

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	1.0168	0.9999	0.8272
Naive	-	NO	1.1444	1.1664	1.0536
Naive	-	YES	1.0209	1.0648	0.9896
Proposal	known	NO	0.9944	1.0217	0.8903
Proposal	known	YES	1.0369	1.0483	0.8909
Proposal	unknown	NO	1.0157	1.0315	0.9063
Proposal	unknown	YES	0.9841	1.0029	0.9324

Table 4: Summary of campaign results 2

(a) Operation period

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	30.0	30.0	30.0
Naive	-	NO	30.0	30.0	24.4
Naive	-	YES	30.0	30.0	29.2
Proposal	known	NO	30.0	30.0	30.0
Proposal	known	YES	30.0	30.0	30.0
Proposal	unknown	NO	30.0	30.0	30.0
Proposal	unknown	YES	30.0	30.0	30.0

(b) Infeasible ratio

Method	Function	Quantile	CPC: 0.6−	CPC: 0.6	CPC: 0.8
True	-	-	-	-	-
Naive	-	NO	0.000	0.053	0.000
Naive	-	YES	0.033	0.047	0.000
Proposal	known	NO	0.027	0.013	0.000
Proposal	known	YES	0.000	0.000	0.000
Proposal	unknown	NO	0.500	0.447	0.020
Proposal	unknown	YES	0.220	0.220	0.020

optimization problems often became infeasible. It was confirmed that introducing quantile regression can mitigate violations of the CPC constraint.

Second, at the 0.8 times CPC setting, budget constraints became the primary bottleneck. Compared to **Naive**, **Proposal** achieved costs closer to the budget limit without premature exhaustion, while simultaneously generating more clicks. This stems from **Proposal**'s ability to regulate the budget consumption pace by incorporating past results as constants into the optimization problem.

6 Conclusion

This paper addressed a previously unexplored setting that uses multiple CPC targets adapting to non-stationary environmental changes during a campaign. We proposed a heuristic optimization algorithm to handle the high uncertainty of ad performance and demonstrated its effectiveness for budget constraints by simulation experiments.

The proposed method still faces challenges in handling CPC constraints. To address this issue, the prediction phase requires a technique that accurately captures CPC characteristics by considering the relationship between clicks and costs. Furthermore, in the optimization phase, since setting constraint thresholds stricter than actual values allows for conservative operation, methods for appropriately adjusting the thresholds are also considered effective.

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